

JSPL EAF Tap-to-Tap Industrial AI Decision Support System

Final Technical Report — Release 1.0

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Executive Summary

This system integrates a frozen Phase 19 stacking regressor, Phase 20.2 physics-guided optimizer, Phase 31 planning-safe Optimizer V2, and Phase 32 hybrid trust framework into a unified FastAPI + Next.js platform for JSPL EAF tap-to-tap decision support. Release 1.0 focuses on production readiness, operator explainability, plant validation, and thesis/publication deliverables — not model retraining.

System Components

Component	Phase	Role	Status
Prediction Engine	19	TTT forecasting	Production
Optimizer	20.2	Recipe optimization	Production
Optimizer V2	31	Planning-safe search	Research mode
Hybrid Engine	32	Trust & consensus	Integrated
Explainability	29	SHAP + narratives	Integrated
Validation	33	Plant result tracking	Integrated

Key Performance Indicators

- Test MAE: ~2.1–3.1 min (Phase 19 hold-out)
- Default recipe (120 t) prediction: ~39.9 min
- Hybrid Reliability Index: 68–78 / 100 on live HMI heats
- Optimizer V2: POWER field never modified

Research Pipeline (Frozen)

Phase 16: Complete — artifacts frozen, no retraining in Release 1.0
Phase 17: Complete — artifacts frozen, no retraining in Release 1.0
Phase 18: Complete — artifacts frozen, no retraining in Release 1.0
Phase 19: Complete — artifacts frozen, no retraining in Release 1.0
Phase 20: Complete — artifacts frozen, no retraining in Release 1.0
Phase 21: Complete — artifacts frozen, no retraining in Release 1.0
Phase 22: Complete — artifacts frozen, no retraining in Release 1.0
Phase 23: Complete — artifacts frozen, no retraining in Release 1.0
Phase 24: Complete — artifacts frozen, no retraining in Release 1.0
Phase 25: Complete — artifacts frozen, no retraining in Release 1.0
Phase 26: Complete — artifacts frozen, no retraining in Release 1.0
Phase 27: Complete — artifacts frozen, no retraining in Release 1.0
Phase 28: Complete — artifacts frozen, no retraining in Release 1.0
Phase 29: Complete — artifacts frozen, no retraining in Release 1.0
Phase 30: Complete — artifacts frozen, no retraining in Release 1.0
Phase 31: Complete — artifacts frozen, no retraining in Release 1.0
Phase 32: Complete — artifacts frozen, no retraining in Release 1.0
Phase 33: Complete — artifacts frozen, no retraining in Release 1.0

Limitations

- Actual TTT for recent live heats pending MES import
- Digital twin sensor completeness gaps (Phase 27)
- Operator feedback does not trigger model retraining